

# Christian Garry

Machine Learning Researcher | Stochastic Control · Deep Learning · Multi-Agent RL

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## Research

### Deep BSDEs for Mean-Field Market Making Durham University

Durham, United Kingdom  
2026 – MSc Dissertation

- Built deep backward-SDE-with-jumps solvers (PyTorch) for a competitive mean-field market-making game (Cont–Xiong); matched the exact solution to 0.135% spread error and scaled to  $\sim 5,000$  agents.
- **Tinkered architectures into working models:** diagnosed and fixed three failure modes that silently kill the mean-field signal (generator bypass, BatchNorm erasure, DeepSets collapse); post-fix the learned competition factor varies  $4\times$  and quotes shift  $2\times$  (26-experiment validation).
- 20-seed multi-agent-RL (MADDPG) study: learned spreads reach **supra-cartel** ( $t=3.22$ ,  $p=0.0022$ , surviving Bonferroni & Holm) — collusion is a property of the learning dynamics, not the market equilibrium.
- Developed an a-posteriori error-certification theory for deep-BSDE solvers (computable estimator, two-sided guarantees, certified stop rule); core machine-checked in Lean 4 / Mathlib ( $\sim 36k$  LOC, axiom-clean).

### Silicon Carbide JFET CPU MEng Dissertation (84%)

Durham, United Kingdom  
Oct 2023 – Apr 2024

- Designed and simulated a 4-bit CPU in SiC JFET logic (LTspice) and built the full verification toolchain (C++ compiler, assembler, Python automation, emulator-parity + waveform-to-state checks).

## Education

### MSc Scientific Computing & Data Analysis (AI for Engineering) Durham University

Durham, United Kingdom  
Sep 2025 – Exp. Sep 2026

- Tracking Distinction — average  $\sim 86\%$  (95% performance/GPU, 94% ML & statistics,  $\sim 89\%$  advanced Bayesian ML); coverage: Bayesian inference, deep learning, convex optimisation (duality, KKT), GPU/HPC.

### Mathematics Modules University of Illinois Urbana-Champaign

NetMath, Online  
Apr 2026 – Dec 2026

- **MATH 314** Higher Mathematics (*in progress*, 100%) & **MATH 447** Real Analysis (*upcoming*) — proof techniques, metric spaces, compactness, Riemann integration.

### MEng Electronic Engineering — Upper Second-Class Honours (2:1) Durham University

Durham, United Kingdom  
Sep 2020 – Jun 2024

## Experience

### Graduate Communications Engineer Siemens PLC

Hebburn, United Kingdom  
Sep 2024 – Present

- Built a Python RAG platform from scratch (local LLMs, vector search) over large technical corpora — cut engineer time-to-answer by  $>90\%$ .
- Implemented C/C++ communication components (TCP/IP, serial) for digital-twin simulation.

### Industrial Tutor Durham University, Dept. of Engineering

Durham, United Kingdom  
Sep 2025 – Present

- Lead weekly design tutorials, mentoring engineering teams on feasibility-driven decisions and giving structured feedback on assumptions and justification quality.

## Technical Skills & Certifications

Deep learning (PyTorch, deep-BSDE/FBSNN, CNNs/Transformers) • architecture & feature debugging • multi-agent RL (MADDPG) • Bayesian inference (MCMC, Gaussian processes) • stochastic control (HJB $\leftrightarrow$ BSDE, mean-field games) • hypothesis testing with multiple-comparison correction • GPU/CUDA & HPC • Python, C++, C, R • Lean 4 / Mathlib

- **MITx 15.455x** — Mathematical Methods for Quantitative Finance (*in progress*); **Columbia** — Introduction to Financial Engineering & Risk Management.